Routes to Safety

An evacuation-services map from open geospatial data — datasets, tiles, routing, rendering

DuckDB · Overture Maps · Planetiler · PMTiles · MapLibre GL JS · OSRM

Vishal Kumar · Shaishav Maisuria

FOSS4G NA 2026, Sacramento — Monday 2 November, 09:00–12:00, Room Beavis

github.com/vishal4c/routes-to-safety-workshop

What we work with — open datasets and open tools

OPEN DATASETS

OpenStreetMap
ODbL

the road network we route on — one Geofabrik .osm.pbf extract

Overture Maps
ODbL / CDLA-P 2.0

roads, places, towns, water, land cover, buildings — five themes, staged as GeoJSON

CAL FIRE FRAP
CC-BY

Fire Hazard Severity Zones and fire perimeters (ArcGIS feature services)

ESA WorldCover
CC-BY 4.0

land cover at 10 m, via the Overture base theme

OPEN TOOLS — one per phase

DuckDB ✦ spatial
Phase 1

SQL over GeoJSON, and over Overture Parquet on S3 — a county out of the planet in seconds

Planetiler → PMTiles
Phase 2a

GeoJSON → vector tiles → one static file served by HTTP Range requests

OSRM
Phase 2b, 2c, 3c

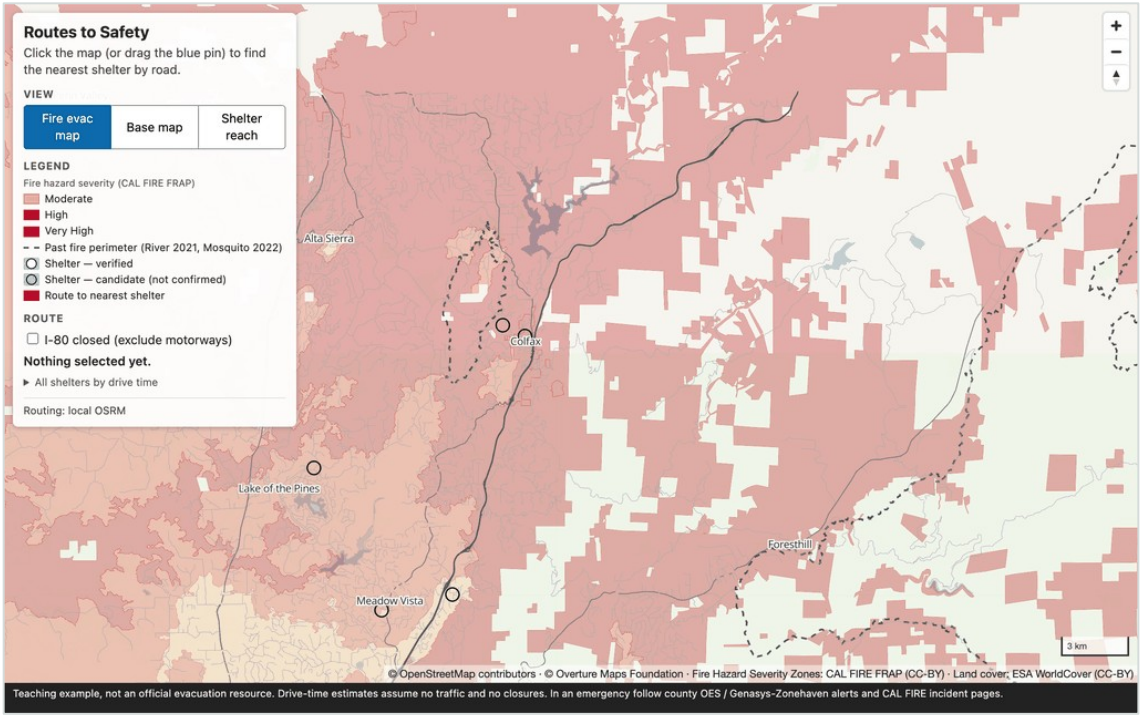
routing graph from the OSM extract; /nearest, /table, /route → nearest by road, reach bands, closures

MapLibre GL JS
Phase 3

draws the tiles from a JSON style — three views, routing wired into the page

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The app in motion



Animates in slideshow mode. Recorded against the local OSRM container; the public demo routes via router.project-osrm.org, without the closure toggle.

1 Drop a pin at Dutch Flat → Sierra Vista CC, 18 min

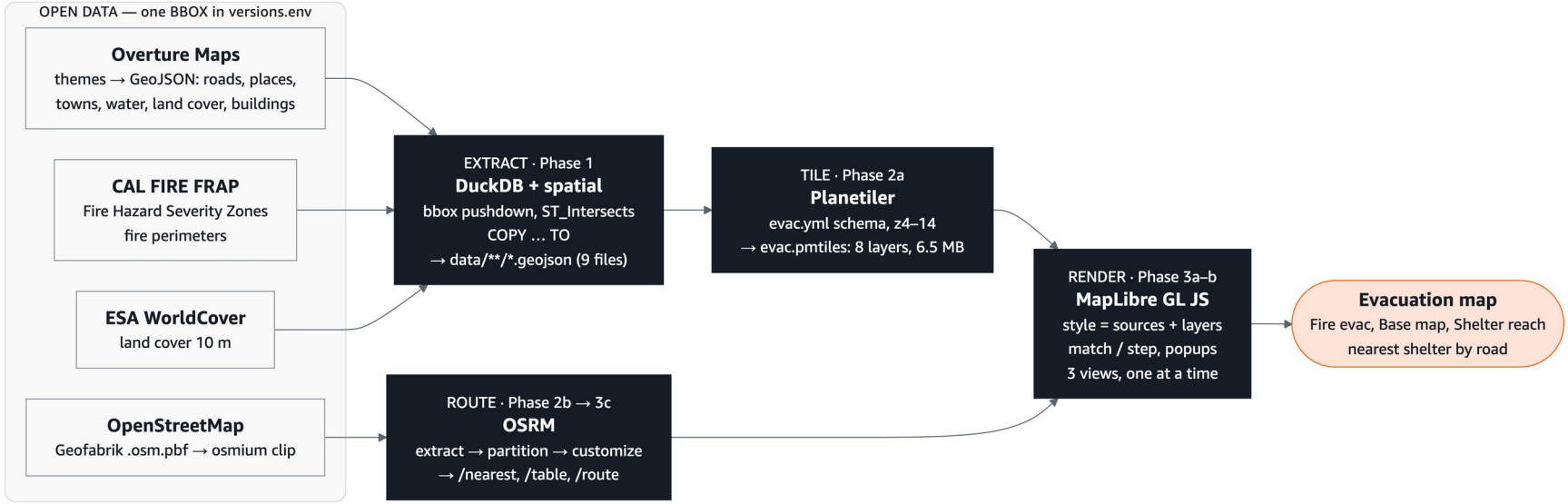
2 Tick "I-80 closed" → same shelter, 31 min via Dutch Flat Rd

3 Foresthill: nearest on the map (Georgetown, 12 km) ≠ nearest by road (Placer High, 36 min)

4 Shelter reach view → 10 / 20 / 30-min bands; click Placer High → only its reach

5 Base map view → a plain street map; the pin and the route stay — routing works in every view

The pipeline: one extract ⟶ a basemap and live routes



Phase 4 change one bbox line in versions.env ⟶ run the nine scripted steps ⟶ the same map for a town of our choice

Pinned	DuckDB 2.0.0	Planetiler 0.10.2	pmtiles 1.31.2	MapLibre GL JS 6.14	OSRM v26.10.0	verified by CHECKSUMS.sha256

Agenda — five phases, one map

0 **Setup**
Pick a way to run, then one checkpoint line — the same green line on every screen
DuckDB

1 **Data prep**
Explore the staged data, pick candidate shelters against the hazard, export GeoJSON
DuckDB · Overture Maps

Break

2 **Map artifacts**
Build the tiles; the routing graph and the reach bands are prebuilt
Planetiler · PMTiles · OSRM

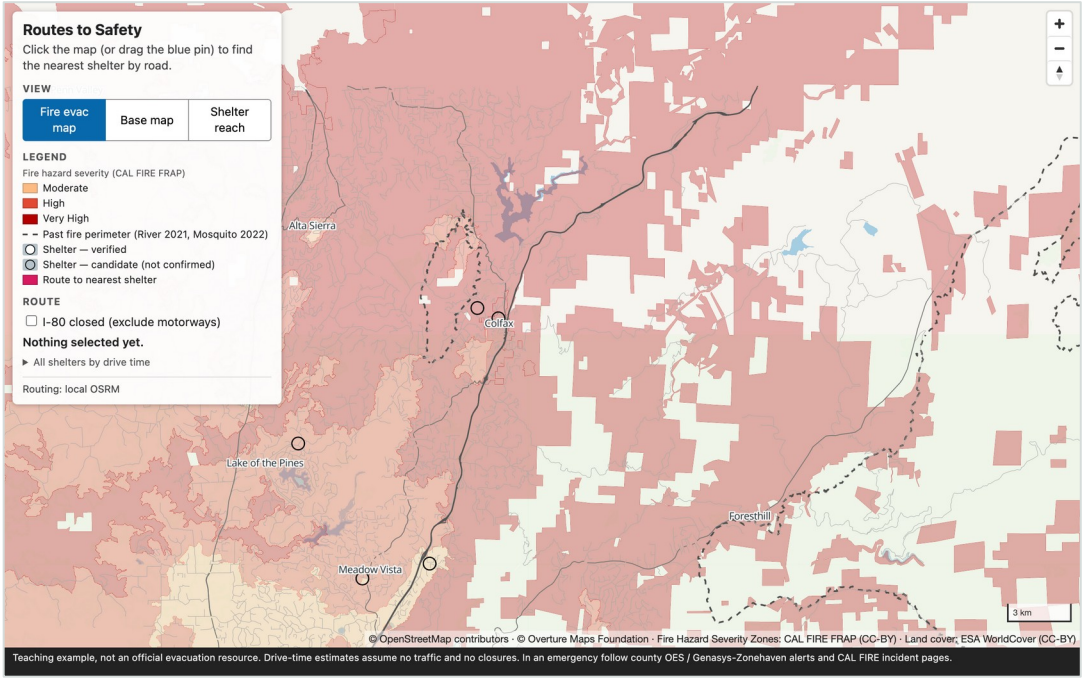
3 **Application**
Style the map, switch between three views, wire in routing, close I-80
MapLibre GL JS · OSRM

4 **Adapt**
One bbox in versions.env, the same nine steps for a town of our choice
versions.env · scripts/

Sticky note on the lid: **green** = fine, **red** = stuck, come find us. Finished early? Every phase card has an extra exercise.

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Study area: the Placer County I-80 corridor



Shelters: 1 verified (Sierra College) + 9 candidates from Overture places. Hazard: CAL FIRE FHSZ 2024 (SRA + LRA). Teaching example, not an official evacuation resource. Drive-time estimates assume no traffic and no closures.

Rocklin → Auburn → Colfax → Dutch Flat / Foresthill

- bbox `−121.40, 38.72, −120.52, 39.20` (W, S, E, N) — about 0.4 square degrees
- 36,294 drivable road segments, 10 shelters

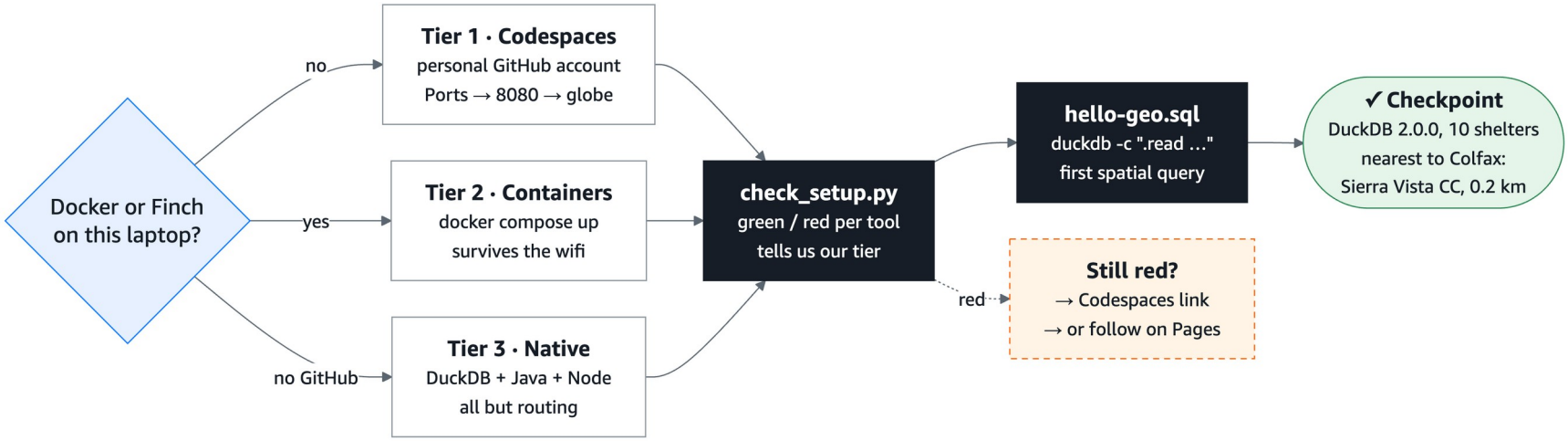
Why here

- River Fire, Aug 2021 — Colfax evacuated for ~2 days
- Mosquito Fire, Sep 2022 — the shelter moved mid-incident to Sierra College: the safe place is not a fixed point
- One interstate, steep canyons — nearest on the map is often not nearest by road

What we leave out

- Shelter status — open, closed or full
- Drive time ≠ evacuation time
- Live road closures
- Shelter capacity

Phase 0 in one picture — pick a way to run, run two lines, hands up



Phase 0 Docker or Finch on the laptop? → Codespaces, containers or native → two commands → hands up

✓ **Checkpoint** OK — DuckDB 2.0.0 · 10 shelters (1 verified, 9 candidate) · nearest to Colfax (crow-flies): Sierra Vista Community Center, 0.2 km

Three ways to run — pick one, keep a neighbour

Tier	Who	Routing	Notes
1 Codespaces	no Docker on the laptop	yes	**personal** GitHub account; sit next to a Docker user port 8080 via the Ports tab ⟶ globe icon
2 Local containers	**recommended** where Docker / Finch is available	yes	survives wifi loss corporate laptops: `finch compose up`
3 Native	DuckDB + Java + Node, no containers	no	Windows: Git Bash map + shelters only — OSRM ships no native Windows / macOS binary
4 Browser only	laptop failed	no	Pages site: follow along; the demo routes via the public OSRM server

PHASE 0 · setup checkpoint

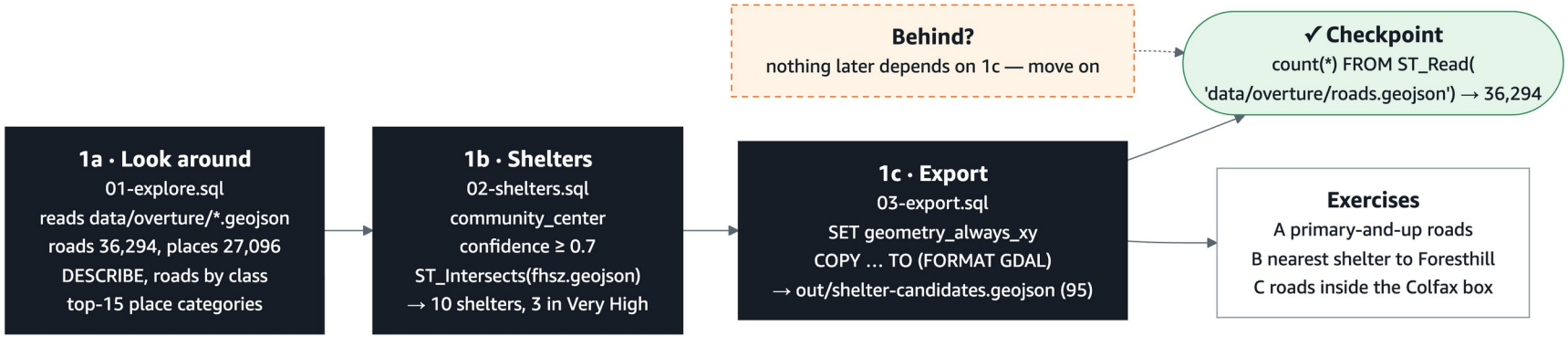
Run both lines from the repo root; hands up when the output matches the line on the right.

Still red? Codespaces link. Blocked? Pages link, follow along.

```
python3 setup/check_setup.py
duckdb -c ".read setup/hello-geo.sql"
```

```
OK — DuckDB 2.0.0 · 10 shelters (1 verified, 9 candidate)
  · nearest to Colfax (crow-flies): Sierra Vista Community Center, 0.2 km
```

Phase 1 in one picture — three SQL files, one folder of GeoJSON



Phase 1 1a explore → 1b candidate shelters → 1c export

✓ **Checkpoint** SELECT count(*) FROM ST_Read('data/overture/roads.geojson') → **36,294**

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Overture Maps: five themes, one box



transportation

drivable road segments

36,294

roads



places

points of interest, with category and confidence

27,096

places



base

water and land cover

1,920 ✦ 2,564

water, land cover



buildings

footprints, height, floors

12,606

buildings (Auburn box)



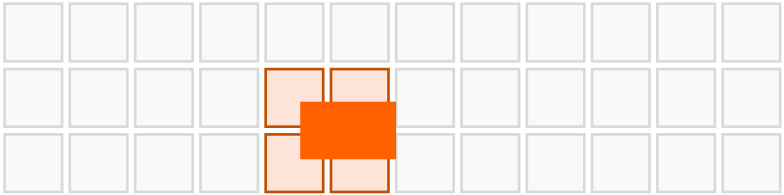
divisions

localities → town labels

121

towns

How a county comes out of the planet in 24 s



every row carries a bbox → DuckDB reads only the row groups our box touches; nothing is downloaded

Linux Foundation project

monthly GeoParquet release on S3 (we pin 2026-09-23.1)

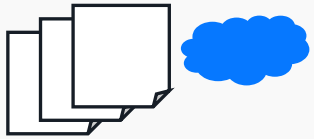
Licence by theme

ODbL from OpenStreetMap, CDLA-Permissive 2.0 places, CC-BY 4.0 land cover

Attribution on every map

© OpenStreetMap contributors, © Overture Maps Foundation

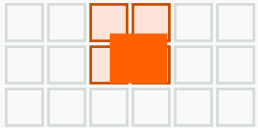
DuckDB: one engine for all the data work



Reads files where they are

GeoJSON, Parquet, CSV: on disk or on S3, no import step

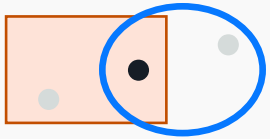
```
ST_Read('roads.geojson')
read_parquet('s3://.../*')
```



Prunes the planet to a box

every Parquet row carries a bbox; only the row groups our box touches are read

```
WHERE bbox.xmin > W
  AND bbox.xmax < E ...
```



Asks spatial questions in SQL

a predicate is a yes/no question about two shapes, in WHERE like any other condition

```
ST_Intersects(a, b)
ST_Distance(a, b)
```



Writes the files the map needs

one COPY writes GeoJSON for Planetiler and MapLibre

```
COPY (...) TO 'x.geojson'
  (FORMAT GDAL)
```

```
LOAD spatial;
SELECT s.name FROM shelters s, fhsz h
WHERE ST_Intersects(s.geom, h.geom) AND h.FHSZ_Description = 'Very High';
```

Phase 1b in three lines: which shelters sit in a Very High zone? Three names come back.

In the room one binary · no server · no account

Pinned DuckDB 2.0.0 · spatial · httpfs

No GIS desktop in this phase: tables in, tables out, one predicate.

Columns cheat sheet — raw Overture vs. our GeoJSON

We want	Raw Overture (S3)	Our data/overture/*.geojson
a place's name	names.primary	name
a place's category	taxonomy.primary	category
how sure Overture is	confidence (use >= 0.7)	same
road type	class, subclass	same
the shape	geometry	same

scripts/fetch-overture.sh already flattened these — today we type the right-hand column.

categories was removed in Sep 2026 → taxonomy.primary

There is no "fairground" category — Gold Country Fairgrounds hides under another one.

Writing GeoJSON: SET geometry_always_xy = true first, or lat/lon come out swapped.

The noise is the lesson: John L Sullivan Chevrolet is a community_center at confidence 0.92.

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PHASE 1 · hands-on — data prep with DuckDB

1a · Look around

`duckdb -c ".read 00-data/01-explore.sql"`

three queries — DESCRIBE roads, roads by class, top-15 place categories; the expected output is printed in the file

1b · Candidate shelters

`duckdb -c ".read 00-data/02-shelters.sql"`

community_center + confidence ≥ 0.7 ⟶ spot the Chevrolet dealer ⟶ open data/shelters.geojson ⟶ ST_Intersects with FHSZ

1c · Export

write GeoJSON with SQL (right):

`duckdb -c ".read 00-data/03-export.sql"` ⟶ out/shelter-candidates.geojson, 95 places

Nothing later depends on 1c — behind? move on.

```
LOAD spatial; SET geometry_always_xy = true;
COPY (SELECT id, name, category, confidence, geom
      FROM ST_Read('data/overture/places.geojson')
      WHERE category IN ('school','community_center', … ))
TO 'out/shelter-candidates.geojson' (FORMAT GDAL, DRIVER 'GeoJSON');
```

Exercises

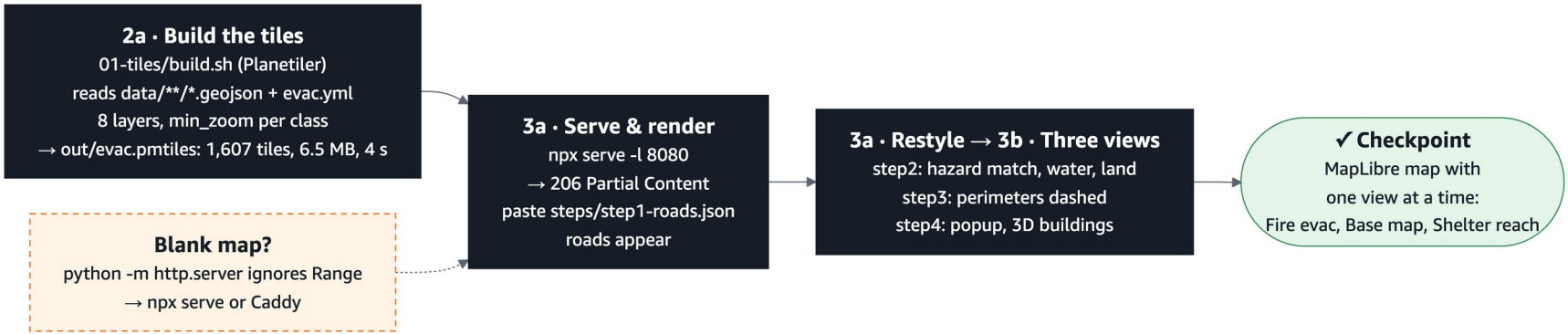
A count motorway + trunk + primary ⟶ 1,909

B crow-flies nearest shelter to Foresthill (−120.8182, 39.0199) ⟶ Georgetown School, 12.4 km. **Remember it for Phase 3c.**

C (finish early) count the roads inside the Colfax box with ST_Intersects

✓ **Checkpoint** `duckdb -c "LOAD spatial; SELECT count(*) FROM ST_Read('data/overture/roads.geojson')"` ⟶ **36,294**

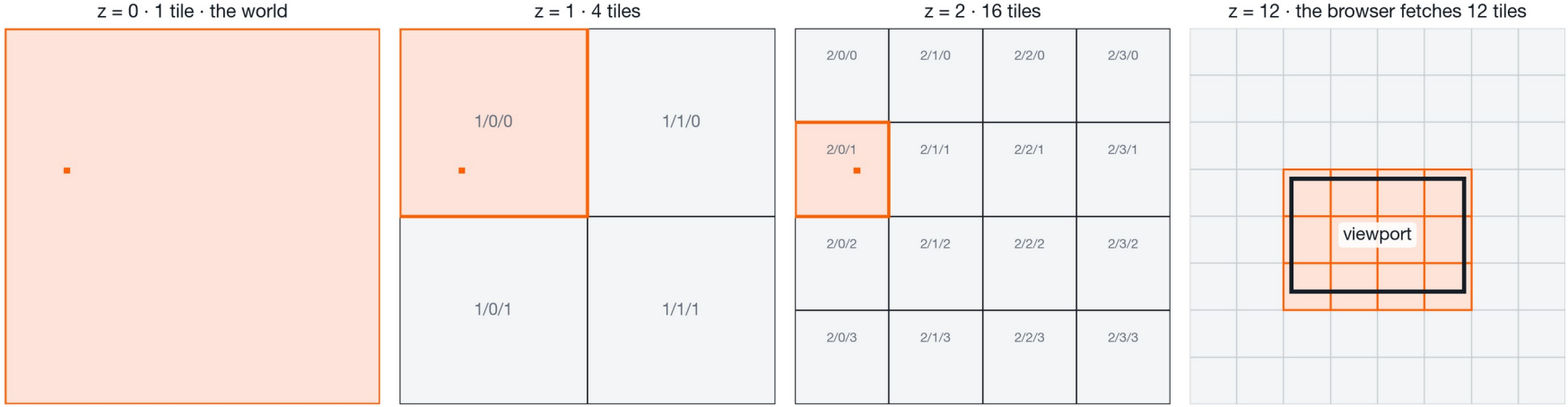
Phase 2 ⟶ 3 in one picture — GeoJSON ⟶ one tile file ⟶ a styled map



Phase 2a ⟶ 3b 2a build the tiles ⟶ 3a serve, render, restyle ⟶ 3b three views

✓ **Checkpoint** curl -sI -H "Range: bytes=0-15" http://localhost:8080/out/evac.pmtiles ⟶ **206 Partial Content**

Why tiles: the map is cut into a z/x/y quadtree



- **Each zoom splits every tile in four.** A tile is addressed z/x/y and holds only what falls inside it.
- **The browser asks for the tiles under the viewport** at the zoom it shows, never the whole dataset.
- **Our file: 1,607 tiles, z4–14, 6.5 MB** instead of a 17 MB GeoJSON and a database.

z	tiles on Earth	over our box
4	256	1
8	65,536	4
10	1,048,576	12
14	268 M	1,218

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Vector tiles, PMTiles and the Range-request trap

Vector tiles hold geometry and attributes, not pixels — the browser draws them, so a restyle needs no rebuild.

PMTiles = every tile in one file with an index. The browser asks for byte ranges: no tile server, any static host (GitHub Pages, S3, a USB stick).

The trap: `python -m http.server` ignores Range headers ⟶ the map stays blank. Use Caddy (:8080) or `npx serve -l 8080 .` and look for **206 Partial Content**.

```
# the only test needed
curl -sI -H "Range: bytes=0-15" http://localhost:8080/out/evac.pmtiles | head –1
HTTP/1.1 206 Partial Content
```

```
# 01-tiles/evac.yml  (excerpt — 8 layers, all local files, zero network)
sources:
  roads: { type: geojson, local_path: data/overture/roads.geojson }
  towns: { type: geojson, local_path: data/overture/towns.geojson }
layers:
  - id: roads
    features:
      - { source: roads, geometry: line, min_zoom: 6,
          include_when: { class: [motorway, trunk, primary, secondary] } }
      - { source: roads, geometry: line, min_zoom: 10, min_size: 1,
          exclude_when: { class: [motorway, trunk, primary, secondary] } }
```

MapLibre style: sources, layers, and two expressions

```
"fill-color": ["match", ["get", "FHSZ_Description"],
  "Moderate",  "#fdbb84",
  "High",      "#e34a33",
  "Very High", "#b30 000",
  "#cccccc"]                    // fallback
```

```
"line-width": ["step", ["zoom"], 0.6, 10, 1.2, 13, 1.6]
```

`match` — one output per category (hazard class ⟶ OrRd)

`step` — one output per range (zoom ⟶ width)

Case and spelling must match exactly: "Very High", not "very_high". FHSZ is also there as 1 | 2 | 3.



A style is JSON: `sources` (our one PMTiles file) + `layers` (one per thing drawn, bottom to top).

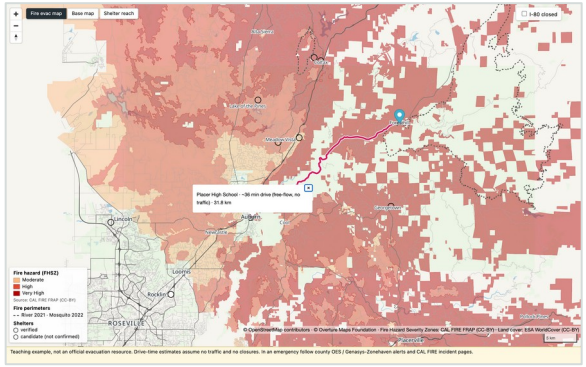
Paste-in steps: index.html sorts pasted layers into a fixed order and de-duplicates by id — paste order never matters.

Labels: a symbol layer with text-field + text-font; glyphs vendored in 02-web/fonts, so labels work offline.

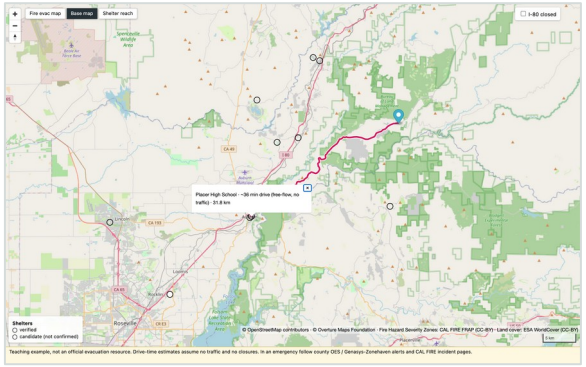
Three views, one at a time: setLayoutProperty(visibility) over a list of layer ids, legend swapped with it — no style reload, so routing keeps working.

Shelter reach: a GeoJSON source + setFilter(shelter_id); the 10 / 20 / 30-min rings never overlap, so they stay translucent over the roads. Precomputed in Phase 2c.

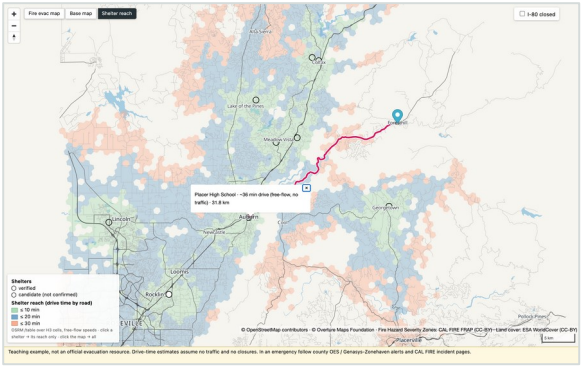
What Phase 3 produces: three views of the same tiles



Fire evac map — our tiles: FHSZ Moderate / High / Very High, dashed perimeters, city labels, shelters, route



Base map — a standard street raster (Esri, no API key); only shelters, the pin and the route on top (needs internet)



Shelter reach — 10 / 20 / 30-minute bands (green, blue, light red) from every shelter; click a shelter for its own bands

Strictly one view at a time

- the legend follows the view; reach bands are never drawn over the hazard fill
- hazard edges are zone boundaries, not a guarantee — a road just outside a polygon is not "safe"

✓ **Phase 3a checkpoint**

- Very High zones solid red, water blue
- 10 shelter circles, 1 filled
- Mosquito perimeter dashed, city labels on
- attribution visible bottom-right

PHASE 2a ⟶ 3b · hands-on — build and style the map

Open the page

Codespaces Ports tab ⟶ 8080 ⟶ globe

Docker / Finch http://localhost:8080/02-web/

Native `npx serve -l 8080 .` ⟶ same URL

2a · Build the tiles

`01-tiles/build.sh` `# = java -jar planetiler.jar generate-custom --schema=01-tiles/evac.yml --output=out/evac.pmtiles`

`pmtiles show out/evac.pmtiles`

Expected: 1,607 tiles, zoom 4–14, 8 layers, ≈ 6.5 MB, ~4 s

No PMTiles file? ⟶ `cp data/prebuilt/evac.pmtiles out/`

3a · Serve & render (step 1)

in 02-web/index.html find

`// STEP 1: paste here` ⟶ paste steps/step1-roads.json ⟶ reload; uncomment AttributionControl ⟶ grey roads + attribution bottom-right

3a · Restyle (steps 2–4)

step2: hazard match (OrRd), water fill, land cover, city labels; shelters filled = verified, hollow = candidate

step3 (if time): fire perimeters dashed grey

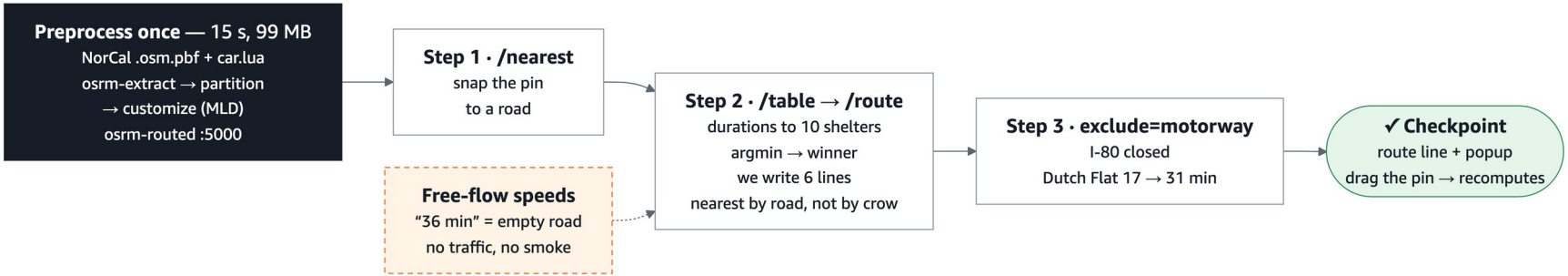
step4 (finish early): click popup with the name; fill-extrusion on buildings (zoom ≥ 13, tilt)

stuck? ⟶ paste steps/step3-styled.json; totally lost ⟶ open 02-web/solution.html

✓ **Checkpoint** Very High solid red, water blue, 10 shelters (1 filled), Mosquito perimeter dashed, attribution visible

3b · Three views — try the Base map and Shelter reach buttons, one view at a time: a plain street map, or each shelter's 10 / 20 / 30-min drive-time bands (click a shelter to isolate it); then come back, it is the same page.

Phase 3c in one picture — preprocess once, then three HTTP calls

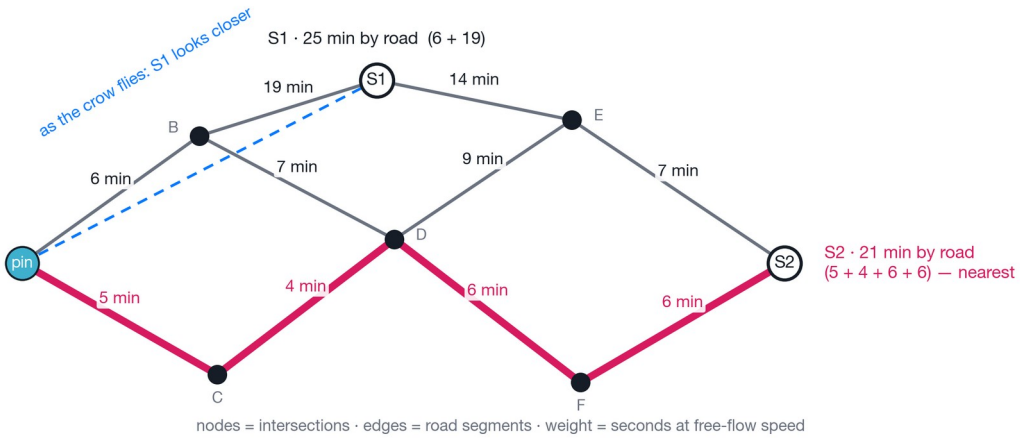


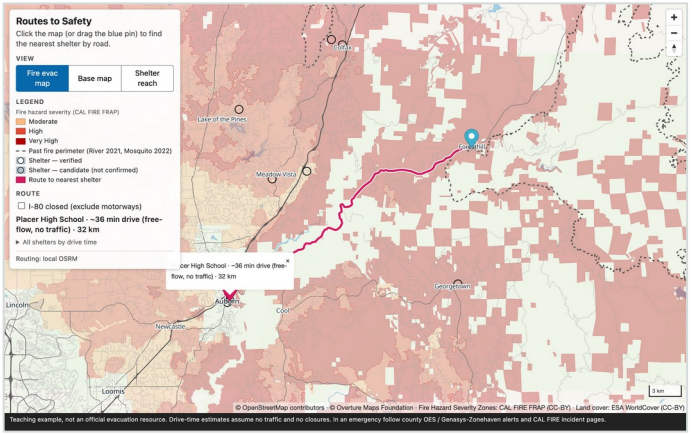
Phase 3c hello route (/nearest) → nearest by road (/table → argmin → /route) → I-80 closed (exclude=motorway)

Teaching numbers

Dutch Flat → Sierra Vista CC: 17.1 min, 31.3 with I-80 closed	Foresthill → Placer High: 35.8 min, 31.8 km

Why a routing graph: nearest on the map ≠ nearest by road





Foresthill: the crow says Georgetown School, 12 km; the road says Placer High, 35.8 min. A canyon lies between.

The graph

nodes are intersections, edges are road segments, each weighted in seconds at free-flow speed

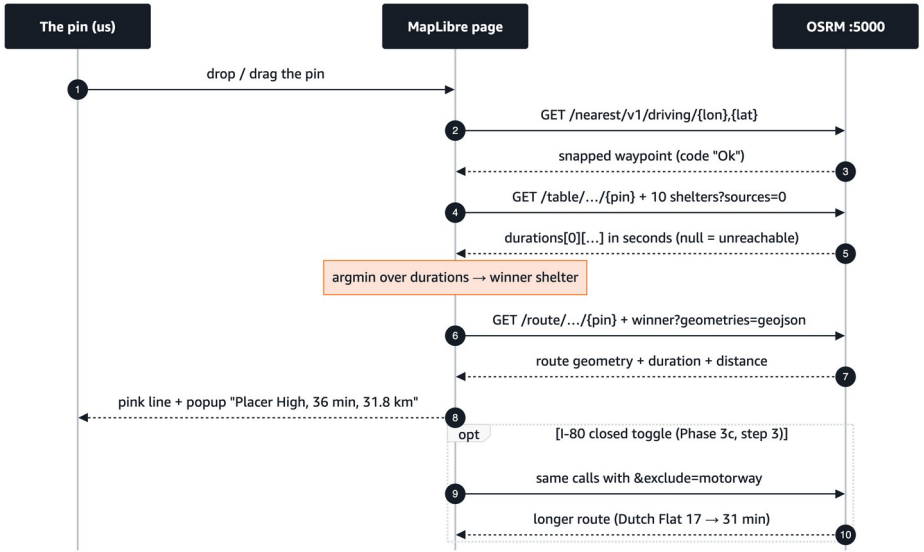
Dijkstra

grow the cheapest frontier out from the pin until a shelter is settled: that shelter is nearest by road

OSRM: preprocess once

extract → partition → customize, 15 s and 99 MB for our box; every call after that answers in milliseconds

Routing is three HTTP calls



Snap, table, argmin, route: the whole of "nearest shelter by road" in one exchange. The six lines we write are the argmin and the /route call; the optional part is the same calls with `exclude=motorway`.

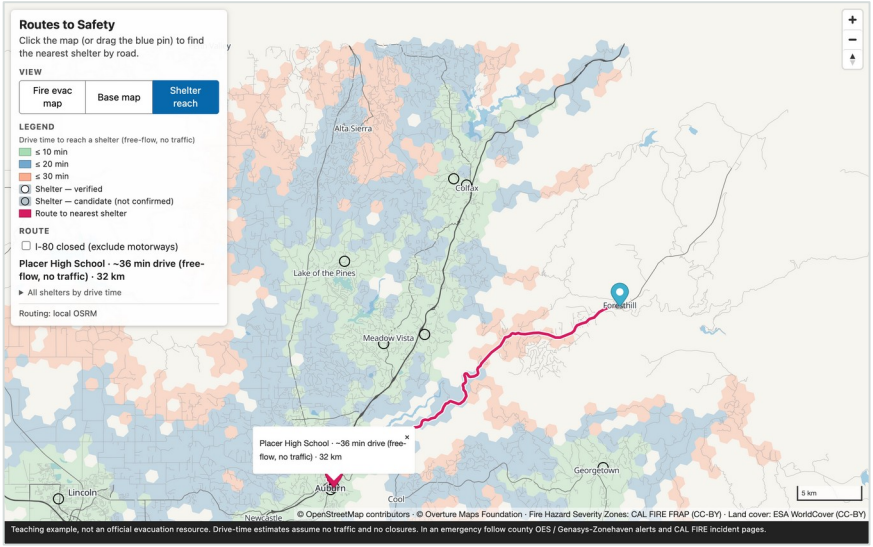
Call	Question it answers
/nearest	which road point is closest to this coordinate? (snap)
/route	path, duration and distance between two points
/table	durations from many points to many points

```
GET $OSRM_URL/{service}/v1/driving/
    {lon},{lat};{lon},{lat}      # longitude first!
```

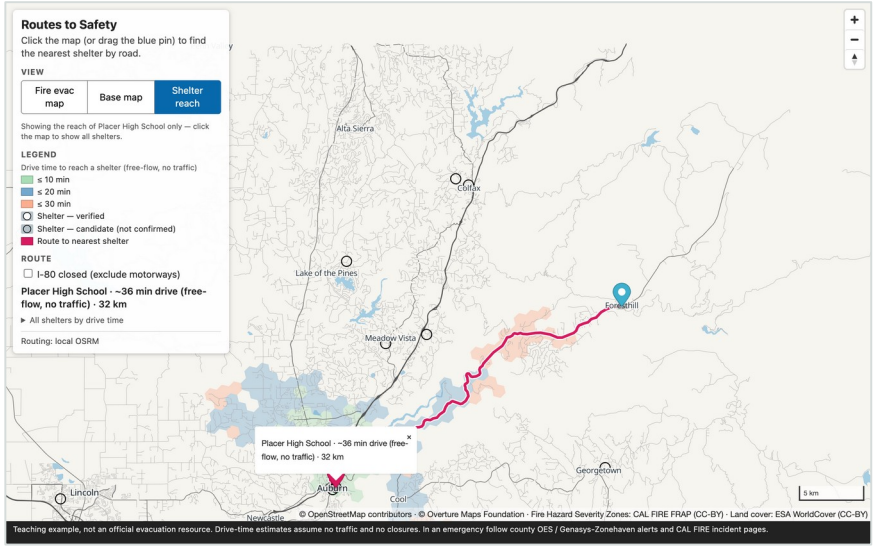
Weights are free-flow speeds, not traffic. "36 minutes" means 36 minutes on an empty road.

Tier	$OSRM_URL
1 Codespaces	<forwarded-8080-URL>/osrm (Ports → globe; never forward 5000)
2 Docker / Finch	http://localhost:8080/osrm (Caddy proxies to osrm:5000)
3 Native, 4 Browser	no routing: pair with a neighbour

From routes to coverage: /table once, draw the reach



Shelter reach view, all shelters — 10 / 20 / 30-minute bands in green, blue, light red (dissolved H3 cells)



Click a shelter ⟶ only its 10 / 20 / 30-minute reach

`scripts/make-coverage.py`: H3 res-8 grid over the bbox (3,341 cells with a road) ⟶ one batched `/table` call (cells × 10 shelters) ⟶ DuckDB dissolves the cells into one band per shelter. Stretch A in Phase 3c is the 400-point version with `grid.sql`.

PHASE 3c · hands-on — routing

Step 1 · Hello route

`curl "$OSRM_URL/nearest/v1/driving/-121.08,38.90"` (PowerShell: curl.exe)

Expected: JSON with "code":"Ok" and a waypoints array. Then /route…? geometries=geojson; paste via steps/step1-route.js → pink line Foresthill → Placer High, console "Route: 35.8 min, 31.8 km".

Step 2 · Nearest shelter by road

in steps/step2-scaffold.js

we write 6 lines: /table URL → argmin durations[0] → /route → setData → popup. Compare with Exercise B. Drag either marker.

not routing? → open steps/step3-solution.js

Step 3 · I-80 is closed

add the

`&exclude=motorway` toggle; start from **Dutch Flat (-120.838, 39.198)**: 17 → 31 min to Sierra Vista CC, the route leaves I-80 for Dutch Flat Rd / Rollins Lake Rd.

Real closures aren't a parameter — nobody flags a downed power line before someone drives into it. This shows the calculation, not detection.

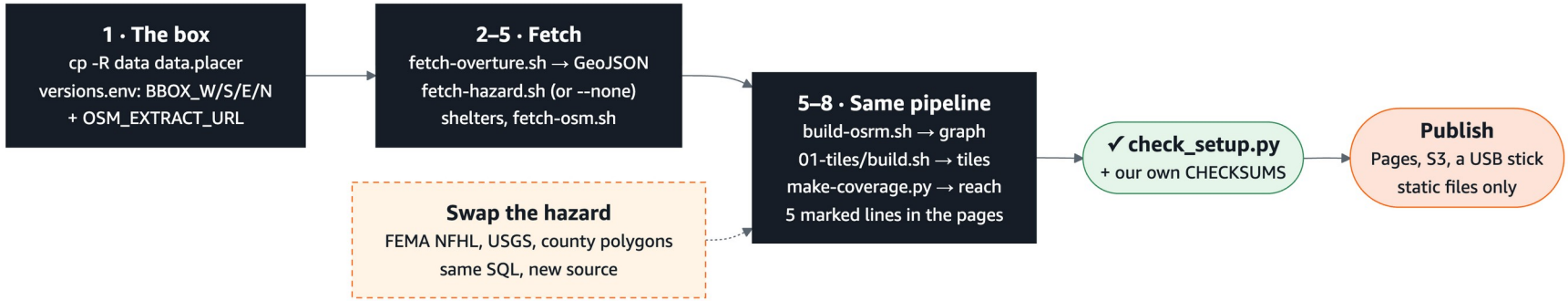
Tiers 3/4: no routing by design — pair with a Docker/Codespaces neighbour or follow the screen.

Finish early? Stretch A — 15-min reach: grid.sql → one /table POST (400 points)

Stretch B — build the graph for Colfax; finish at home, the repo has the solution

✓ **Checkpoint** drag anywhere → line + popup update; the toggle changes the route; footer disclaimer visible

Phase 4 in one picture — one box, the same scripts, our own town



Phase 4 fetch-overture.sh → fetch-hazard.sh → shelters → fetch-osm.sh + build-osrm.sh → build.sh → make-coverage.py → 5 marked lines

Pinned DuckDB 2.0.0 | Planetiler 0.10.2 | pmtiles 1.31.2 | MapLibre GL JS 6.14 | OSRM v26.10.0

Reproducibility, then swap the community

If we can't rebuild it next year with the same answers, it's a demo, not a method.

`versions.env` pins every tool and image; `scripts/check-versions.sh` fails CI on drift

Dated extracts: Overture 2026-09-23.1, Geofabrik norcal 2026-10-22

`data/CHECKSUMS.sha256` verifies the 143 MB bundle we downloaded

One script per step — GeoJSON in, artifacts out:

fetch-overture.sh | fetch-hazard.sh | fetch-osm.sh

build-osrm.sh | 01-tiles/build.sh | make-coverage.py

PHASE 4 · Swap the community

1. cp -R data data.placer (keeps Placer), then in versions.env set BBOX_W/S/E/N and OSM_EXTRACT_URL

2. scripts/fetch-overture.sh (20–40 s), then scripts/fetch-hazard.sh (--none outside California)

3. shelters ⟶ fetch-osm.sh + build-osrm.sh ⟶ build.sh ⟶ make-coverage.py

The OSM region download is the big one (100 MB – 1 GB); everything else runs in seconds.

Swap the hazard

FEMA flood zones instead of fire zones: polygons ⟶ ST_Intersects ⟶ tiles ⟶ match on a zone field. A data change, not a code change.

Going live (CAL FIRE perimeters feed) **raises the stakes on every caveat** we leave out: status, drive time, closures, capacity.

Laptop ⟶ planet: same shape, bigger inputs

Stage	Today (0.4 sq°)	At scale
Extract	DuckDB bbox pushdown, 24 s, six GeoJSON files (38 MB)	same SQL, a region at a time, run where the data lives
Tiles	Planetiler 4 s ⟶ 6.5 MB PMTiles	planet-scale builds are a documented Planetiler use case
Serve	npx serve / Caddy on :8080	PMTiles on object storage behind a CDN — still one file, no tile server
Routing	OSRM MLD graph 15 s, 99 MB	one graph per region; Valhalla / GraphHopper add isochrones and traffic
Freshness	pinned release 2026-09-23.1	Overture publishes monthly — bump the pin, rerun, diff the answers

In the wild

- OSM underlies OSRM — public server at router.project-osrm.org
- Planetiler builds OpenMapTiles-style and custom tile sets
- PMTiles is the Protomaps single-file format
- Overture releases monthly under the Linux Foundation
- MapLibre is the open fork lineage of Mapbox GL
- Real systems add live closures, shelter status, capacity and traffic. **We did not.**

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Recap — the pipeline we built



Open data ✦ open tools: no accounts, no licences to buy, no server to rent — all of it re-runs on a laptop.

Two questions, one extract: where are the safe places, and how do people reach them in time — a basemap and routes from the same data.

It answers them on a good day. It cannot tell anyone if today is a good day: shelter status, traffic, live closures and capacity are not in it.

Keep: the repo (MIT code, CC BY-SA 4.0 materials), the 143 MB data bundle (release data-v2), the public demo, and the habit: one box, one script per step.

Licences, attribution, links

What	Licence
Code	MIT
Materials (slides, docs)	CC BY-SA 4.0
OpenStreetMap	ODbL
Overture Maps	ODbL / CDLA-Permissive 2.0 (by theme)
CAL FIRE FHSZ, perimeters	CC-BY
ESA WorldCover	CC-BY 4.0
Noto Sans glyphs	SIL OFL 1.1

Repo github.com/vishal4c/routes-to-safety-workshop

Demo vishal4c.github.io/routes-to-safety-workshop/demo/

Data bundle .../releases/tag/data-v2 (143 MB; scripts/get-bundle.sh)

Feedback form [URL / QR to add]

Learn more

- duckdb.org/docs/extensions/spatial
- docs.overturemaps.org
- github.com/onthegomap/planetiler
- docs.protomaps.com/pmtiles
- maplibre.org
- project-osrm.org/docs

Attribution to keep on every map

- © OpenStreetMap contributors
- © Overture Maps Foundation
- Base map: © Esri, HERE, Garmin, USGS
- Fire Hazard Severity Zones: CAL FIRE FRAP (CC-BY)
- Land cover: ESA WorldCover (CC-BY)

Thank you.

Vishal Kumar and Shaishav Maisuria — questions: stay, or find us in the hallway.

Teaching example, not an official evacuation resource. Drive-time estimates assume no traffic and no closures.

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